Module Name: CS6351 – Physical Computer

Final Project: Smart Wear

Project Title: Guardian Range Wearable

Student Name: Christian K Anene

Student ID: **25063456**

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Module Instructor: Nicholas ward

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1. ABSTRACT:

This project presents **Guardian Range Wearable**, a smart safety wristband designed to increase awareness of personal safety in public spaces. The device uses an ultrasonic distance sensor, LED lights, and a small LCD screen to show visual and text-based warnings. It is controlled by an Arduino and powered by a portable LiPo battery. The wristband has three alert modes: **green** for safe, **sky blue** for scanning, and **red** for danger. Each mode uses light patterns and scrolling messages to show how close an object is to the wearer.

Inspired by Critical Design, this project goes beyond basic functionality and explores how technology can influence how people feel about safety, space, and danger. This report explains the idea behind the project, the creative design process, related work, and how the system was built and programmed.

2. INTRODUCTION:

Personal safety is an important issue in modern city life, especially for people who move through public spaces alone. Many wearable devices today provide live data, alerts, or health tracking, but they rarely make users think about how safety feels or how technology changes the way we understand danger. **The Guardian Range Wearable** explores these ideas by combining practical technology with critical design. Instead of only detecting nearby objects, the device uses distance to represent personal safety. It alerts the wearer when their personal space is safe, when someone or something is getting close, and when that space has been crossed.

This project asks an important question: what if a wearable device did more than detect danger? What if it helped people become more aware of their surroundings and depend on technology for safety, their sense of safety, and their control over personal space?

3. Background and Related Work

3.1 Wearable Technology and Safety

Many existing safety wearables, such as panic-button bracelets, personal alarms, and GPS tracking devices, are designed to be used only during emergencies. Their main purpose is to send alerts or contact authorities or trusted people. While these tools can be useful, they mostly focus on reacting to danger after it happens. They also often suggest that danger is always present in everyday life, without encouraging users to think about how safety is experienced or felt.

3.2 Critical and Speculative Design

Critical Design, introduced by Dunne and Raby, encourages designers to question common assumptions and explore alternative ways of thinking through design objects. The **Guardian Range Wearable** follows this approach by using a working safety device to explore the emotional and psychological aspects of personal safety. The three light modes **green**, **sky-blue**, and **red**, create a constant background awareness rather than a sudden alarm. The scrolling messages such as "**SAFE FOR NOW**," "**SCANNING… STAND BY,**" and "**YOUR SAFETY IS COMPROMISED**" are intentionally dramatic. They turn everyday changes in distance into moments for reflection, asking users to think about how much fear and attention we place on possible danger.

3.3 Ambient Display Research

Ambient displays use gentle visual signals instead of loud or disruptive alerts. This project is inspired by research in Calm Technology, ambient lighting, and proximity-based visual feedback. The **LED** behaviours soft green breathing, flowing blue gradients, and flashing red light follow ideas from ambient computing. At the same time, the effects are designed to be expressive and noticeable, highlighting how technology can both calm and intensify our awareness of safety.

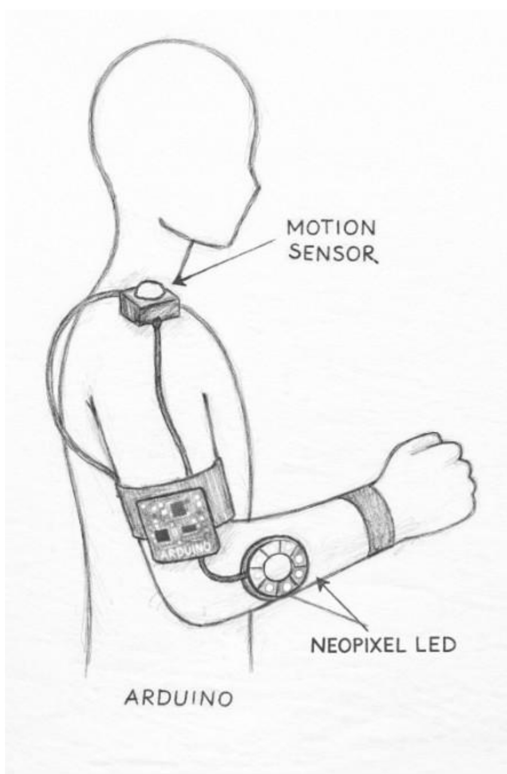
4. Creative Process:

The creative process of the **Guardian Range Wearable** developed in three main stages: exploring the idea, building and testing prototypes, and refining the final wearable form.

4.1 Conceptual Exploration and Concept Sketching

At the beginning, the project focused on creating a wearable device for personal safety. Over time, this idea shifted toward a critical design approach. Instead of trying to reduce fear, the project asked whether fear itself could be made visible. This change influenced both the messages shown on the LCD and how the LEDs behaved. The color system, green for permission or safety, blue for uncertainty, and red for warning was inspired by common security systems such as airport alerts and emergency signs that people already recognize and react to.

Early sketches examined different body placements (wrist, arm, shoulder). The wrist was chosen for its visibility and symbolic proximity to pulse/heartbeat.



4.2 Mapping Emotions to Colors

- **Red** = danger (high urgency)
- **Sky blue** = scanning (neutral assessment)
- **Green** = safe (calm reassurance)

4.3 Iterative Prototyping

The prototyping stage involved testing different LED animations to see how they affected emotional response. A slow, breathing green light felt calm and reassuring. A moving blue gradient suggested scanning and uncertainty. A flashing red light clearly communicated urgency and danger. Feedback from testing showed that lighting behavior strongly influenced how serious each alert felt to users.

The LCD messages were also carefully chosen. Messages like "**YOUR SAFETY IS COMPROMISED**" were intentionally dramatic. They exaggerate the meaning of simple distance detection to show how technology can increase feelings of anxiety, even when the actual risk may be low.

4.4 Iteration Through Critical Questions

Throughout the design and prototyping process, a series of critical questions guided the conceptual and creative development of the Smart Safety Wristband. Rather than focusing solely on technical performance or usability, these questions were used as reflective tools to interrogate broader social, emotional, and ethical implications of wearable safety technologies.

How can a wearable prompt user to rethink their **sense of safety**?

The wristband challenges the idea that safety is something clear and measurable. By using colors and short messages to show "safe" or "danger," the device turns a personal feeling into something visible. This makes users stop and think about why they feel unsafe.

It encourages users to ask themselves whether they are in danger or whether their fear comes from news, social media, or living in a culture where danger is often expected. Messages like "**SAFE FOR NOW**" and "**YOUR SAFETY IS COMPROMISED**" are intentionally uncertain. They show that safety is not permanent and can change at any time.

In this way, the wristband reflects personal anxiety back to the wearer and invites them to question how much their sense of safety is shaped by their surroundings and society, rather than real immediate threats.

What does it mean for the body to become a "**sensor platform**"?

By wearing the sensor on the body, the wristband turns the user into part of the sensing system. The body is no longer just being protected by technology, it becomes the place where information is collected and displayed.

This raises questions about control and trust. If the wristband tells the user when danger is present, they may start relying on the device instead of their own instincts. The project explores whether technology helps people become more aware of their surroundings or whether it replaces natural human judgment.

This idea reflects a wider trend in society, where people increasingly use wearable devices to monitor their bodies and environments, allowing technology to shape how they experience the world.

Does technology **reduce fear** or increase it?

Although safety technologies are meant to make people feel safer, this project questions whether they sometimes do the opposite. Constant warning lights and scanning messages can keep users alert all the time, making it feel like danger is always nearby.

The scanning (sky blue) state is important because it represents uncertainty rather than comfort. It shows a world that is always being checked, never fully safe or unsafe. This reflects how modern life often feels, always waiting for something to happen.

Instead of removing fear, the wristband makes it visible. It asks whether technology truly helps people feel safer or whether it makes danger more noticeable and emotionally exhausting.

5. System design and implementation

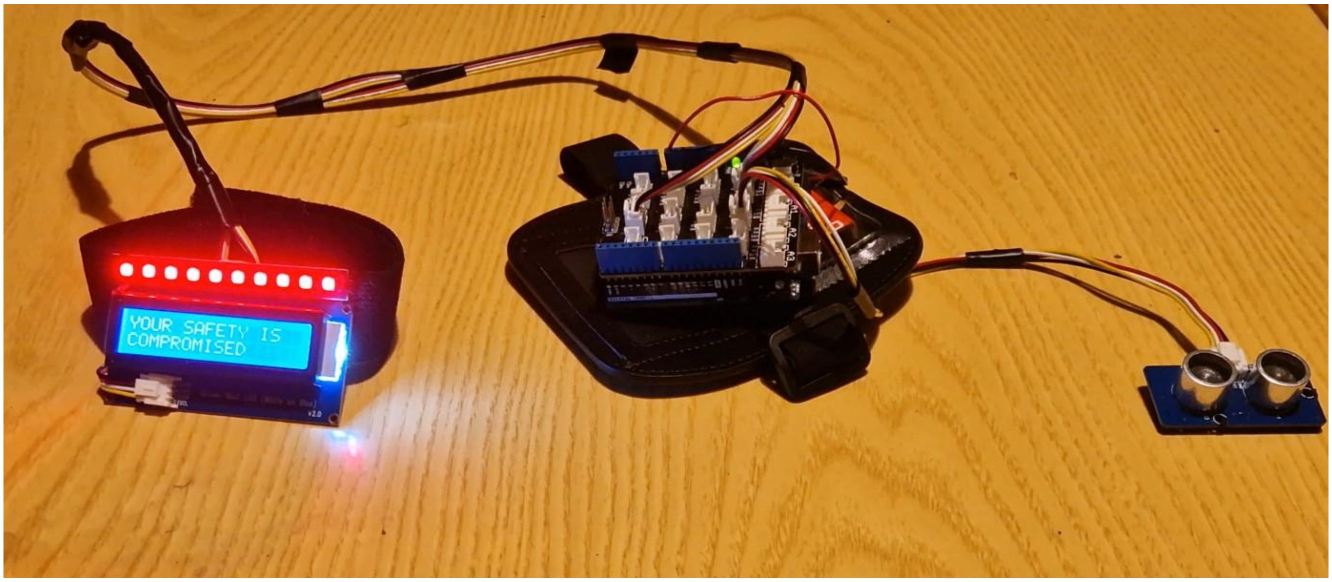
The system uses a single-pin **ultrasonic** sensor for distance measurement, a WS2813 10-LED stick for visual output, and a Grove 16×2 RGB LCD for text messaging. The Arduino runs a mode-based architecture controlled by a **switch statement**.

Mode	LED Behavior	LCD Message	Meaning
0 (Danger)	Red Strobe	"YOUR SAFETY IS - COMPROMISED"	Object too close (<10 cm)
1 (Safe)	Solid green	"SAFE FOR NOW - STAY AWARE"	Clear distance (≥30 cm)
2 (Scanning)	Deep Sky-Blue gradient	"SCANNING… STAND BY"	Uncertain mid-range

5.1 Input: Ultrasonic Sensor

The ultrasonic sensor is used to represent environmental awareness. Technically, it measures distance to nearby objects, but in this project, it has a more symbolic role. It stands in for the idea of detecting possible danger in the environment.

Rather than accurately identifying real threats, the sensor represents how people constantly scan their surroundings for potential risk, both physical and psychological. This reflects how individuals often feel unsafe even when no clear danger is present. The sensor therefore supports the project's goal of exploring how fear and awareness are shaped, rather than providing guaranteed protection.



5.2 Output: Neo Pixel LED Strip

The Neo Pixel LED strip is the main visual communication tool of the wristband. It provides quick, easy-to-understand feedback through color and motion. Each LED state is designed to match a specific emotional response.

- **Red** (**Danger**):
  The red flashing pattern is fast and attention grabbing. It creates a sense of urgency and discomfort, reflecting how danger is often experienced emotionally rather than logically.

- **Sky Blue** (**Scanning**):
  The flowing blue gradient represents uncertainty. It shows that the system is constantly checking the environment, even when no immediate threat is detected. This state reflects the feeling of being alert and unsure, which is common in modern public spaces.

- **Green** (**Safe**):
  The soft breathing effect of the green LEDs communicates calm and temporary reassurance. The slow fade in and out suggests that safety is present, but not permanent.

These LED animations are intentionally designed to mimic emotional states rather than simply show system status. This aligns with critical design principles by encouraging reflection instead of offering clear answers.

5.3 Output: LCD Display (16×2)

The LCD display provides short text messages that support the visual signals from the LEDs. The messages are written in a reflective and speculative tone rather than giving direct instructions.

Examples include:

- "**YOUR SAFETY IS / COMPROMISED**"
- "**SCANNING... / STAND BY**"
- "**SAFE FOR NOW / STAY AWARE**"

The wording is intentionally serious and uncertain. It avoids absolute claims and instead highlights how fragile and temporary safety can be. By combining text with color and animation, the device communicates both emotional and informational meaning.

Together, the LCD messages reinforce the idea that the wristband is not a perfect safety solution, but a tool for thinking about fear, awareness, and reliance on technology.

6. DISCUSSION

The **Guardian Range Wearable** is both a working safety device and a design that makes people think. It can detect how close an object is, but it also asks questions about how much we depend on technology to tell us when we are safe or in danger. By turning distance into strong lights and messages, the device shows that safety is not only about space, but also about how we feel.

People experience safety differently depending on where they live. In countries where guns are allowed and common, many people feel more alert and anxious in public spaces. Someone standing too close can quickly feel dangerous. The red flashing lights and strong warning messages of the Guardian Range Wearable reflect this constant feeling of alertness and fear.

In countries where guns are not allowed, public spaces are often felt as calmer and safer. People may not expect danger from everyday situations. In these places, the wearable's strong warnings can feel extreme. This helps show how technology can increase fear even when the real risk is low.

The wearable also shows how much power technology has over our emotions. When the device says, "**YOUR SAFETY IS COMPROMISED**," the wearer is likely to believe it, even if nothing dangerous is happening. This makes us think about how much we trust devices instead of our own judgment.

The project also highlights the stress of always being monitored. Safety devices are meant to protect us, but they can also make us feel nervous all the time. The Guardian Range Wearable exaggerates this effect to help people notice the hidden fear that many carry in daily life.

Overall, the wearable explores the balance between feeling protected and becoming dependent on technology. It encourages users to think about how safety, fear, and technology are connected, and how culture and laws influence how safe we feel.

7. CONCLUSION

The **Guardian Range Wearable** shows that wearable technology can be more than just useful it can also make people reflect. By using a distance sensor, **LED** lights, and a small screen, the device turns simple measurements into emotional safety signals.

Instead of offering a perfect safety solution, the project asks important questions about fear and trust. It shows how technology can shape how we feel about danger, especially in different countries with different rules about violence and weapons.

Through a Critical Design approach, this project shows that safety devices do not just protect us, they also change how we think and feel. The Guardian Range Wearable invites users to ask: **Does technology truly make us safer, or does it teach us to always expect danger**?

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Thank You